Package ‘manCTMed’

January 19, 2025

Title Continuous Time Mediation

Version 0.9.2

Description Research compendium for the manuscript
Pesigan, I. J. A., Russell, M. A., Chow, S.-M. (In Preparation).
Standard Errors and Confidence Intervals for Total, Direct, and Indirect Effects
in Continuous-Time Mediation Models.
<doi:10.0000/0000000000>.

URL https://github.com/jeksterslab/manCTMed,
https://jeksterslab.github.io/manCTMed/, https://osf.io/qwnmf/,
https://doi.org/10.0000/0000000000

BugReports https://github.com/jeksterslab/manCTMed/issues

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Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

Depends R (>= 4.0.0)

Imports utils, stats, dynr, OpenMx, dynUtils, simStateSpace,
bootStateSpace, cTMed, ggplot2, MASS

Suggests knitr, rmarkdown, testthat, DT

Remotes jeksterslab/dynUtils, jeksterslab/cTMed,
jeksterslab/simStateSpace, jeksterslab/bootStateSpace

RoxygenNote 7.3.2

NeedsCompilation no

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`BootPara` *Parametric Bootstrap*

Description

The function generates simulated datasets based on a fitted model and refits the model to each generated dataset using the `dynr` package.

Usage

```
BootPara(fit, path, prefix, taskid, B = 1000L, ncores = NULL, seed = NULL)
```

Arguments

fit	R object. Output of the FitDynr() function.
path	Path to a directory to store bootstrap samples and estimates.
prefix	Character string. Prefix used for the file names for the bootstrap samples and estimates.
taskid	Positive integer. Task ID.
B	Positive integer. Number of bootstrap samples.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)

## End(Not run)
```

BootParaStdXMY — *Parametric Bootstrap Confidence Intervals for X-M-Y (Standardized)*

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \to M \to Y$ (Standardized).

Usage

```
BootParaStdXMY(boot, theta_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
library(simStateSpace)
library(cTMed)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
theta_hat <- ThetaHat(fit)
ci <- BootParaStdXMY(boot = boot, theta_hat = theta_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

BootParaStdXYM *Parametric Bootstrap Confidence Intervals for X-Y-M (Standardized)*

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \to Y \to M$ (Standardized).

Usage

```
BootParaStdXYM(boot, theta_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
library(simStateSpace)
library(cTMed)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
theta_hat <- ThetaHat(fit)
ci <- BootParaStdXYM(boot = boot, theta_hat = theta_hat)
```

```
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

BootParaStdYMX *Parametric Bootstrap Confidence Intervals for Y-M-X (Standardized)*

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $Y \to M \to X$ (Standardized).

Usage

```
BootParaStdYMX(boot, theta_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
library(simStateSpace)
library(cTMed)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
```

```
  taskid = 1,
  B = 1000L
)
theta_hat <- ThetaHat(fit)
ci <- BootParaStdYMX(boot = boot, theta_hat = theta_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

BootParaXMY *Parametric Bootstrap Confidence Intervals for X-M-Y*

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \to M \to Y$.

Usage

```
BootParaXMY(boot, phi_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
```

```
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
phi_hat <- PhiHat(fit)
ci <- BootParaXMY(boot = boot, phi_hat = phi_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

BootParaXYM *Parametric Bootstrap Confidence Intervals for X-Y-M*

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \to Y \to M$.

Usage

```
BootParaXYM(boot, phi_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
phi_hat <- PhiHat(fit)
ci <- BootParaXYM(boot = boot, phi_hat = phi_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

BootParaYMX *Parametric Bootstrap Confidence Intervals for Y-M-X*

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $Y \to M \to X$.

Usage

```
BootParaYMX(boot, phi_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
boot <- BootPara(
  fit = fit,
  path = getwd(),
  prefix = "pb",
  taskid = 1,
  B = 1000L
)
phi_hat <- PhiHat(fit)
ci <- BootParaYMX(boot = boot, phi_hat = phi_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

`Compress` *Compress Replication*

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

DeltaStdXMY *Delta Method Confidence Intervals for X-M-Y (Standardized)*

Description

The function generates delta method confidence intervals for the mediation model $X \to M \to Y$ (Standardized).

Usage

```
DeltaStdXMY(theta_hat, delta_t = 1:30)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- DeltaStdXMY(theta_hat)
plot(ci)

## End(Not run)
```

DeltaStdXYM *Delta Method Confidence Intervals for X-Y-M (Standardized)*

Description

The function generates delta method confidence intervals for the mediation model $X \to Y \to M$ (Standardized).

Usage

```
DeltaStdXYM(theta_hat, delta_t = 1:30)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- DeltaStdXYM(theta_hat)
plot(ci)

## End(Not run)
```

DeltaStdYMX *Delta Method Confidence Intervals for Y-M-X (Standardized)*

Description

The function generates delta method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$ (Standardized).

Usage

```
DeltaStdYMX(theta_hat, delta_t = 1:30)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- DeltaStdYMX(theta_hat)
plot(ci)

## End(Not run)
```

DeltaXMY *Delta Method Confidence Intervals for X-M-Y*

Description

The function generates delta method confidence intervals for the mediation model $X \to M \to Y$.

Usage

```
DeltaXMY(phi_hat, delta_t = 1:30)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- DeltaXMY(phi_hat)
plot(ci)

## End(Not run)
```

DeltaXYM *Delta Method Confidence Intervals for X-Y-M*

Description

The function generates delta method confidence intervals for the mediation model $X \to Y \to M$.

Usage

```
DeltaXYM(phi_hat, delta_t = 1:30)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- DeltaXYM(phi_hat)
plot(ci)

## End(Not run)
```

DeltaYMX *Delta Method Confidence Intervals for Y-M-X*

Description

The function generates delta method confidence intervals for the mediation model $Y \to M \to X$.

Usage

```
DeltaYMX(phi_hat, delta_t = 1:30)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- DeltaYMX(phi_hat)
plot(ci)

## End(Not run)
```

FigPlotEffects *Plot Total, Direct, and Indirect Effects*

Description

Effects for the model $X \to M \to Y$.

Usage

```
FigPlotEffects(dynamics = 0, std = FALSE, max_delta_t = 30)
```

Arguments

dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.
max_delta_t	Numeric. Maximum time interval.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: FigScatterPlotCoverage(), FigScatterPlotPower(), FigScatterPlotType1()

Examples

```
FigPlotEffects()
```

FigScatterPlotCoverage *Plot Coverage Probabilities*

Description

Coverage probabilities for the model $X \to M \to Y$.

Usage

```
FigScatterPlotCoverage(results, delta_t = NULL, dynamics = 0, std = FALSE)
```

Arguments

results	Summary results data frame.
delta_t	Vector of time-interval value. If delta_t = NULL, use all available time-intervals
dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: FigPlotEffects(), FigScatterPlotPower(), FigScatterPlotType1()

Examples

```
data(results, package = "manCTMed")
FigScatterPlotCoverage(results)
FigScatterPlotCoverage(results, delta_t = 1:14)
FigScatterPlotCoverage(results, delta_t = 15:30)
```

FigScatterPlotPower *Plot Statistical Power*

Description

Statistical power for the model $X \rightarrow M \rightarrow Y$.

Usage

```
FigScatterPlotPower(results, delta_t = NULL, dynamics = 0, std = FALSE)
```

Arguments

results	Summary results data frame.
delta_t	Vector of time-interval value. If delta_t = NULL, use all available time-intervals
dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: FigPlotEffects(), FigScatterPlotCoverage(), FigScatterPlotType1()

Examples

```
data(results, package = "manCTMed")
FigScatterPlotPower(results)
FigScatterPlotPower(results, delta_t = 1:14)
FigScatterPlotPower(results, delta_t = 15:30)
```

FigScatterPlotType1 *Plot Type I Error*

Description

Type I error for the model $Y \rightarrow M \rightarrow X$.

Usage

```
FigScatterPlotType1(results, delta_t = NULL, dynamics = 0, std = FALSE)
```

Arguments

results	Summary results data frame.
delta_t	Vector of time-interval value. If delta_t = NULL, use all available time-intervals
dynamics	Integer. dynamics = 0 for original drift matrix, dynamics = -1 for near-neutral dynamics, and dynamics = 1 for stronger damping.
std	Logical. If std = TRUE, standardized total, direct, and indirect effects. If std = FALSE, unstandardized total, direct, and indirect effects.

Author(s)

Ivan Jacob Agaloos Pesigan

See Also

Other Figure Functions: FigPlotEffects(), FigScatterPlotCoverage(), FigScatterPlotPower()

Examples

```
data(results, package = "manCTMed")
FigScatterPlotType1(results)
FigScatterPlotType1(results, delta_t = 1:14)
FigScatterPlotType1(results, delta_t = 15:30)
```

FitDynr — *Fit the Model using the dynr Package*

Description

The function fits the model using the dynr package.

Usage

```
FitDynr(data, taskid)
```

Arguments

data	R object. Output of the RandomMeasurement() function.
taskid	Positive integer. Task ID.

See Also

Other Model Fitting Functions: FitMx(), Grundy2007FitDynr(), PhiHat(), ThetaHat()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
FitDynr(data, taskid = 1)

## End(Not run)
```

FitMx — *Fit the Model using the OpenMx Package*

Description

The function fits the model using the OpenMx package.

Usage

```
FitMx(data, taskid)
```

Arguments

data	R object. Output of the RandomMeasurement() function.
taskid	Positive integer. Task ID.

See Also

Other Model Fitting Functions: FitDynr(), Grundy2007FitDynr(), PhiHat(), ThetaHat()

Examples

```
## Not run:
set.seed(42)
library(OpenMx)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
FitMx(data, taskid = 1)

## End(Not run)
```

GenData *Simulate Data*

Description

The function simulates data using the simStateSpace::SimSSMOUFixed() function.

Usage

```
GenData(taskid)
```

Arguments

taskid	Positive integer. Task ID.

See Also

Other Data Generation Functions: Grundy2007GenData(), RandomMeasurement()

Examples

```
## Not run:
set.seed(42)
sim <- GenData(taskid = 1)
plot(sim)

## End(Not run)
```

grundy2007 *grundy2007*

Description

grundy2007

Usage

```
data(grundy2007)
```

Format

A dataframe with 4123 rows and 5 columns:

id Participant ID variable.

time Time variable

x Mother-reported marital conflict.

m Maternal knowledge.

y Child-reported behavioral competence.

Author(s)

Ivan Jacob Agaloos Pesigan

References

Grundy, A. M., Gondoli, D. M., & Blodgett Salafia, E. H. (2007). Marital conflict and preadolescent behavioral competence: Maternal knowledge as a longitudinal mediator. Journal of Family Psychology, 21 (4), 675–682. doi:10.1037/08933200.21.4.675

Grundy2007BootPara *Parametric Bootstrap (Grundy, et al. 2007)*

Description

The function generates simulated datasets based on a fitted model and refits the model to each generated dataset using the `dynr` package.

Usage

```
Grundy2007BootPara(
  fit,
  data,
  path,
  prefix,
  B = 1000L,
  ncores = NULL,
  seed = NULL
)
```

Arguments

fit	R object. Output of the FitDynr() function.
data	R object. Output of the Grundy2007GenData() function.
path	Path to a directory to store bootstrap samples and estimates.
prefix	Character string. Prefix used for the file names for the bootstrap samples and estimates.
B	Positive integer. Number of bootstrap samples.
ncores	Positive integer. Number of cores to use.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
GrundyBootPara(
  fit = fit,
  data = data,
  path = getwd(),
  prefix = "pb",
  B = 1000L
)

## End(Not run)
```

Grundy2007BootParaStdXMY

Parametric Bootstrap Confidence Intervals for X-M-Y (Standardized)

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \to M \to Y$ (Standardized).

Usage

```
Grundy2007BootParaStdXMY(boot, theta_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
boot <- GrundyBootPara(
  fit = fit,
  data = data,
  path = getwd(),
  prefix = "pb",
  B = 1000L
)
theta_hat <- ThetaHat(fit)
ci <- Grundy2007BootParaStdXMY(boot = boot, theta_hat = theta_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

Grundy2007BootParaStdXYM

Parametric Bootstrap Confidence Intervals for X-Y-M (Standardized)

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \to Y \to M$ (Standardized).

Usage

```
Grundy2007BootParaStdXYM(boot, theta_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
boot <- GrundyBootPara(
  fit = fit,
  data = data,
  path = getwd(),
  prefix = "pb",
  B = 1000L
)
theta_hat <- ThetaHat(fit)
ci <- Grundy2007BootParaStdXYM(boot = boot, theta_hat = theta_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

Grundy2007BootParaXMY *Parametric Bootstrap Confidence Intervals for X-M-Y*

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \to M \to Y$.

Usage

```
Grundy2007BootParaXMY(boot, phi_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
boot <- GrundyBootPara(
  fit = fit,
  data = data,
  path = getwd(),
  prefix = "pb",
  B = 1000L
)
phi_hat <- PhiHat(fit)
ci <- Grundy2007BootParaXMY(boot = boot, phi_hat = phi_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

Grundy2007BootParaXYM *Parametric Bootstrap Confidence Intervals for X-Y-M*

Description

The function generates parametric bootstrap method confidence intervals for the mediation model $X \to Y \to M$.

Usage

```
Grundy2007BootParaXYM(boot, phi_hat, delta_t = 1:30, ncores = NULL)
```

Arguments

boot	R object. Output of the BootPara() function.
phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
ncores	Positive integer. Number of cores to use.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
boot <- GrundyBootPara(
  fit = fit,
  data = data,
  path = getwd(),
  prefix = "pb",
  B = 1000L
)
phi_hat <- PhiHat(fit)
ci <- Grundy2007BootParaXYM(boot = boot, phi_hat = phi_hat)
plot(ci)
plot(ci, type = "bc")

## End(Not run)
```

Grundy2007DeltaStdXMY *DeltaStd Method Confidence Intervals for Conflict-Knowledge-Competence (Grundy, et al. 2007)*

Description

The function generates delta method confidence intervals for the mediation model conflict-knowledge-competence.

Usage

```
Grundy2007DeltaStdXMY(theta_hat, delta_t = 1:30)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
theta_hat <- ThetaHat(fit)
ci <- Grundy2007DeltaStdXMY(theta_hat)
plot(ci)

## End(Not run)
```

`Grundy2007DeltaStdXYM` *DeltaStd Method Confidence Intervals for Conflict-Competence-Knowledge (Grundy, et al. 2007)*

Description

The function generates delta method confidence intervals for the mediation model conflict-competence-knowledge.

Usage

```
Grundy2007DeltaStdXYM(theta_hat, delta_t = 1:30)
```

Arguments

`theta_hat`	R object. Output of the `ThetaHat()` function.
`delta_t`	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: `BootPara()`, `BootParaStdXMY()`, `BootParaStdXYM()`, `BootParaStdYMX()`, `BootParaXMY()`, `BootParaXYM()`, `BootParaYMX()`, `DeltaStdXMY()`, `DeltaStdXYM()`, `DeltaStdYMX()`, `DeltaXMY()`, `DeltaXYM()`, `DeltaYMX()`, `Grundy2007BootPara()`, `Grundy2007BootParaStdXMY()`, `Grundy2007BootParaStdXYM()`, `Grundy2007BootParaXMY()`, `Grundy2007BootParaXYM()`, `Grundy2007DeltaStdXMY()`, `Grundy2007DeltaXMY()`, `Grundy2007DeltaXYM()`, `Grundy2007MCStdXMY()`, `Grundy2007MCStdXYM()`, `Grundy2007MCXMY()`, `Grundy2007MCXYM()`, `MCStdXMY()`, `MCStdXYM()`, `MCStdYMX()`, `MCXMY()`, `MCXYM()`, `MCYMX()`

Examples

```
## Not run:
set.seed(42)
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
theta_hat <- ThetaHat(fit)
ci <- Grundy2007DeltaStdXYM(theta_hat)
plot(ci)

## End(Not run)
```

Grundy2007DeltaXMY *Delta Method Confidence Intervals for Conflict-Knowledge-Competence (Grundy, et al. 2007)*

Description

The function generates delta method confidence intervals for the mediation model conflict-knowledge-competence.

Usage

```
Grundy2007DeltaXMY(phi_hat, delta_t = 1:30)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
phi_hat <- PhiHat(fit)
ci <- Grundy2007DeltaXMY(phi_hat)
plot(ci)

## End(Not run)
```

Grundy2007DeltaXYM *Delta Method Confidence Intervals for Conflict-Competence-Knowledge (Grundy, et al. 2007)*

Description

The function generates delta method confidence intervals for the mediation model conflict-competence-knowledge.

Usage

```
Grundy2007DeltaXYM(phi_hat, delta_t = 1:30)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
phi_hat <- PhiHat(fit)
ci <- Grundy2007DeltaXYM(phi_hat)
plot(ci)

## End(Not run)
```

Grundy2007FitDynr *Fit the CT-VAR Model (Grundy, et al. 2007)*

Description

Fit the CT-VAR Model (Grundy, et al. 2007)

Usage

```
Grundy2007FitDynr(data)
```

Arguments

data	R object. Output of the Grundy2007GenData() function.

See Also

Other Model Fitting Functions: FitDynr(), FitMx(), PhiHat(), ThetaHat()

Examples

```
## Not run:
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
summary(fit)

## End(Not run)
```

Grundy2007GenData *Generate Data (Grundy, et al. 2007)*

Description

Generate Data (Grundy, et al. 2007)

Usage

```
Grundy2007GenData(taskid, seed = NULL, empirical = FALSE)
```

Arguments

taskid	Positive integer. Task ID.
seed	Integer. Random seed.
empirical	logical. If true, mu and Sigma specify the empirical not population mean and covariance matrix.

See Also

Other Data Generation Functions: GenData(), RandomMeasurement()

Examples

```
## Not run:
Grundy2007GenData(taskid = 1)

## End(Not run)
```

Grundy2007MCStdXMY *Monte Carlo Method Confidence Intervals for Conflict-Knowledge-Competence (Grundy, et al. 2007)*

Description

The function generates Monte Carlo method confidence intervals for the mediation model conflict-knowledge-competence.

Usage

```
Grundy2007MCStdXMY(theta_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
theta_hat <- ThetaHat(fit)
ci <- Grundy2007MCStdXMY(theta_hat, seed = 42)
plot(ci)

## End(Not run)
```

Grundy2007MCStdXYM *Monte Carlo Method Confidence Intervals for Conflict-Competence-Knowledge (Grundy, et al. 2007)*

Description

The function generates Monte Carlo method confidence intervals for the mediation model conflict-competence-knowledge.

Usage

```
Grundy2007MCStdXYM(theta_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
theta_hat <- ThetaHat(fit)
ci <- Grundy2007MCStdXYM(theta_hat, seed = 42)
plot(ci)

## End(Not run)
```

Grundy2007MCXMY *Monte Carlo Method Confidence Intervals for Conflict-Knowledge-Competence (Grundy, et al. 2007)*

Description

The function generates Monte Carlo method confidence intervals for the mediation model conflict-knowledge-competence.

Usage

```
Grundy2007MCXMY(phi_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
phi_hat <- PhiHat(fit)
ci <- Grundy2007MCXMY(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

Grundy2007MCXYM *Monte Carlo Method Confidence Intervals for Conflict-Competence-Knowledge (Grundy, et al. 2007)*

Description

The function generates Monte Carlo method confidence intervals for the mediation model conflict-competence-knowledge.

Usage

```
Grundy2007MCXYM(phi_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
data <- Grundy2007GenData(taskid = 1)
fit <- Grundy2007FitDynr(data)
phi_hat <- PhiHat(fit)
ci <- Grundy2007MCXYM(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCStdXMY *Monte Carlo Method Confidence Intervals for X-M-Y (Standardized)*

Description

The function generates Monte Carlo method confidence intervals for the mediation model $X \rightarrow M \rightarrow Y$ (Standardized).

Usage

```
MCStdXMY(theta_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- MCStdXMY(theta_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCStdXYM *Monte Carlo Method Confidence Intervals for X-Y-M (Standardized)*

Description

The function generates Monte Carlo method confidence intervals for the mediation model $X \rightarrow Y \rightarrow M$ (Standardized).

Usage

```
MCStdXYM(theta_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdYMX(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- MCStdXYM(theta_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCStdYMX *Monte Carlo Method Confidence Intervals for Y-M-X (Standardized)*

Description

The function generates Monte Carlo method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$ (Standardized).

Usage

```
MCStdYMX(theta_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

theta_hat	R object. Output of the ThetaHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCXMY(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
theta_hat <- ThetaHat(fit)
ci <- MCStdYMX(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCXMY *Monte Carlo Method Confidence Intervals for X-M-Y*

Description

The function generates Monte Carlo method confidence intervals for the mediation model $X \rightarrow M \rightarrow Y$.

Usage

```
MCXMY(phi_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXYM(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- MCXMY(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCXYM *Monte Carlo Method Confidence Intervals for X-Y-M*

Description

The function generates Monte Carlo method confidence intervals for the mediation model $X \rightarrow Y \rightarrow M$.

Usage

```
MCXYM(phi_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCYMX()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- MCXYM(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

MCYMX *Monte Carlo Method Confidence Intervals for Y-M-X*

Description

The function generates Monte Carlo method confidence intervals for the mediation model $Y \rightarrow M \rightarrow X$.

Usage

```
MCYMX(phi_hat, delta_t = 1:30, R = 20000L, seed = NULL)
```

Arguments

phi_hat	R object. Output of the PhiHat() function.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.
seed	Integer. Random seed.

See Also

Other Confidence Interval Functions: BootPara(), BootParaStdXMY(), BootParaStdXYM(), BootParaStdYMX(), BootParaXMY(), BootParaXYM(), BootParaYMX(), DeltaStdXMY(), DeltaStdXYM(), DeltaStdYMX(), DeltaXMY(), DeltaXYM(), DeltaYMX(), Grundy2007BootPara(), Grundy2007BootParaStdXMY(), Grundy2007BootParaStdXYM(), Grundy2007BootParaXMY(), Grundy2007BootParaXYM(), Grundy2007DeltaStdXMY(), Grundy2007DeltaStdXYM(), Grundy2007DeltaXMY(), Grundy2007DeltaXYM(), Grundy2007MCStdXMY(), Grundy2007MCStdXYM(), Grundy2007MCXMY(), Grundy2007MCXYM(), MCStdXMY(), MCStdXYM(), MCStdYMX(), MCXMY(), MCXYM()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(taskid = 1)
data <- RandomMeasurement(sim)
fit <- FitDynr(data, taskid = 1)
phi_hat <- PhiHat(fit)
ci <- MCYMX(phi_hat, seed = 42)
plot(ci)

## End(Not run)
```

---

params *Simulation Parameters*

Description

Simulation Parameters

Usage

```
data(params)
```

Format

A dataframe with 30 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

dynamics Dynamics. `0` for original drift matrix, `-1` for near-neutral dynamics, and `1` for stronger damping.

Author(s)

Ivan Jacob Agaloos Pesigan

PhiHat — *Estimated Drift Matrix*

Description

The function extracts the estimated drift matrix from the fitted model.

Usage

```
PhiHat(fit)
```

Arguments

fit	R object. Output of the FitDynr() function.

See Also

Other Model Fitting Functions: FitDynr(), FitMx(), Grundy2007FitDynr(), ThetaHat()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(n = 50)
data <- RandomMeasurement(sim)
fit <- FitDynr(data)
PhiHat(fit)

## End(Not run)
```

RandomMeasurement — *Simulate Random Measurement*

Description

The function randomly selects 100 observations from the generated data and replaces the unselected observations with NA.

Usage

```
RandomMeasurement(sim)
```

Arguments

sim	R object. Output of the GenData() function.

See Also

Other Data Generation Functions: GenData(), Grundy2007GenData()

Examples

```
## Not run:
set.seed(42)
sim <- GenData(taskid = 1)
RandomMeasurement(sim)

## End(Not run)
```

---

results *Simulation Results*

Description

Simulation Results

Usage

```
data(results)
```

Format

A dataframe with 1440 rows and 21 columns:

taskid Simulation Task ID.

replications Number of simulation replications.

output_type Simulation output type.

xmy If TRUE, the mediation model is $X \to M \to Y$. If FALSE, the mediation model is $Y \to M \to X$.

std If TRUE, standardized total, direct, and indirect effects. If FALSE, unstandardized total, direct, and indirect effects.

dynamics Dynamics. 0 for original drift matrix, -1 for near-neutral dynamics, and 1 for stronger damping.

n Sample size.

method Method used to generate confidence intervals.

effect Total, direct, or indirect effect.

interval Time-interval.

est Average parameter estimate.

se Average standard error.

z Average z statistic.

p Average p-value.

R Number of replications for the Monte Carlo method.

2.5% Lower limit of the 95% confidence interval.

97.5% Upper limit of the 95% confidence interval.

parameter Population parameter.

zero_hit Proportion of replications where the confidence intervals contained zero.

theta_hit Proportion of replications where the confidence intervals contained the population parameter.

sig If TRUE, $p < 0.05$.

Author(s)

Ivan Jacob Agaloos Pesigan

`Sim` *Simulation Replication*

Description

Simulation Replication

Usage

```
Sim(taskid, repid, output_folder, overwrite, integrity, delta_t, R)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`delta_t`	Numeric vector. Vector of time intervals.
`R`	Positive integer. Number of random drift matrices to generate.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimDynrDeltaStdXMY` *Simulation Replication - DynrDeltaStdXMY*

Description

Simulation Replication - DynrDeltaStdXMY

Usage

```
SimDynrDeltaStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`delta_t`	Numeric vector. Vector of time intervals.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrDeltaStdYMX *Simulation Replication - DynrDeltaStdYMX*

Description

Simulation Replication - DynrDeltaStdYMX

Usage

```
SimDynrDeltaStdYMX(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrDeltaXMY *Simulation Replication - DynrDeltaXMY*

Description

Simulation Replication - DynrDeltaXMY

Usage

```
SimDynrDeltaXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrDeltaYMX *Simulation Replication - DynrDeltaYMX*

Description

Simulation Replication - DynrDeltaYMX

Usage

```
SimDynrDeltaYMX(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrMCStdXMY *Simulation Replication - DynrMCStdXMY*

Description

Simulation Replication - DynrMCStdXMY

Usage

```
SimDynrMCStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrMCStdYMX *Simulation Replication - DynrMCStdYMX*

Description

Simulation Replication - DynrMCStdYMX

Usage

```
SimDynrMCStdYMX(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrMCXMY *Simulation Replication - DynrMCXMY*

Description

Simulation Replication - DynrMCXMY

Usage

```
SimDynrMCXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimDynrMCYMX *Simulation Replication - DynrMCYMX*

Description

Simulation Replication - DynrMCYMX

Usage

```
SimDynrMCYMX(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitDynr *Simulation Replication - FitDynr*

Description

Simulation Replication - FitDynr

Usage

```
SimFitDynr(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN — *Simulation File Name*

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dynr", "dynr-delta-xmy", "dynr-delta-ymx", "dynr-mc-xmy", and "dynr-mc-ymx"
output_folder	Character string. Output folder.
suffix	Character string. Output of manCTMed:::.SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData — *Simulation Replication - GenData*

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimGrundy2007` *Simulation Replication - SimGrundy2007*

Description

Simulation Replication - SimGrundy2007

Usage

```
SimGrundy2007(
  taskid,
  repid,
  output_folder,
  overwrite,
  integrity,
  delta_t,
  R,
  B
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`delta_t`	Numeric vector. Vector of time intervals.
`R`	Positive integer. Number of random drift matrices to generate.
`B`	Positive integer. Number of bootstrap samples.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007BootPara *Simulation Replication - Grundy2007BootPara*

Description

Simulation Replication - Grundy2007BootPara

Usage

```
SimGrundy2007BootPara(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  B
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
B	Positive integer. Number of bootstrap samples.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007BootParaStdXMY

Simulation Replication - Grundy2007BootParaStdXMY

Description

Simulation Replication - Grundy2007BootParaStdXMY

Usage

```
SimGrundy2007BootParaStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007BootParaStdXYM

Simulation Replication - Grundy2007BootParaStdXYM

Description

Simulation Replication - Grundy2007BootParaStdXYM

Usage

```
SimGrundy2007BootParaStdXYM(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007BootParaXMY

Simulation Replication - Grundy2007BootParaXMY

Description

Simulation Replication - Grundy2007BootParaXMY

Usage

```
SimGrundy2007BootParaXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007BootParaXYM

Simulation Replication - Grundy2007BootParaXYM

Description

Simulation Replication - Grundy2007BootParaXYM

Usage

```
SimGrundy2007BootParaXYM(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007DynrDeltaStdXMY

Simulation Replication - Grundy2007DynrDeltaStdXMY

Description

Simulation Replication - Grundy2007DynrDeltaStdXMY

Usage

```
SimGrundy2007DynrDeltaStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007DynrDeltaStdXYM
Simulation Replication - Grundy2007DynrDeltaStdXYM

Description

Simulation Replication - Grundy2007DynrDeltaStdXYM

Usage

```
SimGrundy2007DynrDeltaStdXYM(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007DynrDeltaXMY

Simulation Replication - Grundy2007DynrDeltaXMY

Description

Simulation Replication - Grundy2007DynrDeltaXMY

Usage

```
SimGrundy2007DynrDeltaXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007DynrDeltaXYM

Simulation Replication - Grundy2007DynrDeltaXYM

Description

Simulation Replication - Grundy2007DynrDeltaXYM

Usage

```
SimGrundy2007DynrDeltaXYM(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007DynrMCStdXMY

Simulation Replication - Grundy2007DynrMCStdXMY

Description

Simulation Replication - Grundy2007DynrMCStdXMY

Usage

```
SimGrundy2007DynrMCStdXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007DynrMCStdXYM
Simulation Replication - Grundy2007DynrMCStdXYM

Description

Simulation Replication - Grundy2007DynrMCStdXYM

Usage

```
SimGrundy2007DynrMCStdXYM(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007DynrMCXMY

Simulation Replication - Grundy2007DynrMCXMY

Description

Simulation Replication - Grundy2007DynrMCXMY

Usage

```
SimGrundy2007DynrMCXMY(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`delta_t`	Numeric vector. Vector of time intervals.
`R`	Positive integer. Number of random drift matrices to generate.

Details

This function is executed via the `SimGrundy2007` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007DynrMCXYM

Simulation Replication - Grundy2007DynrMCXYM

Description

Simulation Replication - Grundy2007DynrMCXYM

Usage

```
SimGrundy2007DynrMCXYM(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity,
  delta_t,
  R
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.
delta_t	Numeric vector. Vector of time intervals.
R	Positive integer. Number of random drift matrices to generate.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007FitDynr *Simulation Replication - Grundy2007FitDynr*

Description

Simulation Replication - Grundy2007FitDynr

Usage

```
SimGrundy2007FitDynr(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimGrundy2007GenData *Simulation Replication - Grundy2007GenData*

Description

Simulation Replication - Grundy2007GenData

Usage

```
SimGrundy2007GenData(
  taskid,
  repid,
  output_folder,
  seed,
  suffix,
  overwrite,
  integrity
)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the SimGrundy2007 function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimProj — *Simulation Project Name*

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

SumHit — *Summarize Simulations - CI Hit*

Description

Summarize Simulations - CI Hit

Usage

```
SumHit(taskid, reps, output_folder, output_type, ncores)
```

Arguments

`taskid`	Positive integer. Task ID.
`reps`	Positive integer. Number of replications.
`output_folder`	Character string. Output folder.
`output_type`	Character string. Output type. Valid values include `"data"`, `"fit-dynr"`, `"dynr-delta-xmy"`, `"dynr-delta-ymx"`, `"dynr-mc-xmy"`, and `"dynr-mc-ymx"`
`ncores`	Positive integer. Number of cores to use.

Value

Returns a numeric matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

SumRaw — *Summarize Simulations - Raw*

Description

Summarize Simulations - Raw

Usage

```
SumRaw(taskid, reps, output_folder, output_type, ncores)
```

Arguments

taskid	Positive integer. Task ID.
reps	Positive integer. Number of replications.
output_folder	Character string. Output folder.
output_type	Character string. Output type. Valid values include "data", "fit-dynr", "dynr-delta-xmy", "dynr-delta-ymx", "dynr-mc-xmy", and "dynr-mc-ymx"
ncores	Positive integer. Number of cores to use.

Value

Returns a numeric matrix.

Author(s)

Ivan Jacob Agaloos Pesigan

ThetaHat — *Estimated Drift Matrix and Process Noise*

Description

The function extracts the estimated drift matrix and process noise from the fitted model.

Usage

```
ThetaHat(fit)
```

Arguments

fit	R object. Output of the FitDynr() function.

See Also

Other Model Fitting Functions: FitDynr(), FitMx(), Grundy2007FitDynr(), PhiHat()

Examples

```
## Not run:
set.seed(42)
library(dynr)
sim <- GenData(n = 50)
data <- RandomMeasurement(sim)
fit <- FitDynr(data)
ThetaHat(fit)

## End(Not run)
```

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